

Handwritten Digit Recognition: A Comparative Analysis of Custom SVM and Deep Learning Architectures

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Abstract—Handwritten digit recognition is a complex challenge in computer vision due to the significant intra-class variability of human handwriting. This paper presents a systematic, pipeline-driven comparative study of two fundamentally distinct mathematical approaches to classifying the MNIST dataset: a classical Support Vector Machine (SVM) utilizing a custom hybrid kernel enhanced by Principal Component Analysis (PCA), and a hierarchical Convolutional Neural Network (CNN) operating on raw spatial data. Rather than evaluating isolated models, we construct a continuous four-notebook experimental pipeline spanning data preprocessing, baseline evaluation, hyperparameter optimization, and robustness stress-testing. A rigorous ablation study across three SVM kernel configurations demonstrates that a custom hybrid kernel—blending the global geometric mapping of a Polynomial kernel with the localized precision of a Radial Basis Function (RBF)—maximizes the SVM’s discriminative capacity on PCA-compressed features. Concurrently, we evaluate a deep learning pipeline, transitioning from an overfitting basic CNN to an advanced architecture incorporating dual convolutional blocks, MaxPooling, and Dropout regularization, ultimately achieving a test accuracy of 99.29%. Finally, a stress-test using additive Gaussian noise ($\sigma = 0.3$) quantifies model degradation under real-world visual corruption. The study empirically confirms that while the optimized SVM offers highly competitive accuracy and interpretable decision boundaries, the CNN demonstrates significantly better spatial understanding and noise resilience, making it more suitable for practical vision applications.

Index Terms—Handwritten Digit Recognition, Support Vector Machine, Hybrid Kernel, Convolutional Neural Network, Principal Component Analysis, MNIST Dataset, t-SNE, Robustness Evaluation.

I. INTRODUCTION

The classification of handwritten numerals into distinct categories (0 through 9) is a foundational benchmark in the evolution of artificial intelligence. It serves as the mathematical backbone for numerous real-world applications, including automated postal mail routing, high-speed bank cheque processing, and the digitization of historical archives. The primary challenge in this domain lies in the immense structural variability of human handwriting. Digits of the same class often exhibit drastically different stroke thicknesses, rotational slants, stylistic loops, and spatial translations, rendering rigid template-matching algorithms insufficient for robust classification.

The MNIST dataset [7], established by LeCun et al. [1], serves as the canonical environment for this research. Com-

prising 70,000 grayscale images (28×28 pixel resolution), the dataset is meticulously balanced and complex enough to expose the limitations of basic linear classifiers, while remaining computationally accessible for rapid prototyping.

This paper investigates the dichotomy between classical machine learning and modern deep learning by evaluating two distinct paradigms. The first approach relies on Support Vector Machines (SVM). Because standard SVMs require images to be flattened into 1D vectors—removing their inherent 2D spatial relationships—we must rely heavily on explicit feature engineering (PCA) and complex mathematical transformations (Custom Kernels) to separate the data. The second approach leverages Convolutional Neural Networks (CNN), which natively ingest 2D matrices and automatically learn to detect spatial hierarchies (edges, curves, and complete digits).

Our comparative analysis focuses on three core objectives: (1) quantifying the impact of training data volume on algorithmic convergence, (2) providing a mathematical justification for custom kernel design in high-dimensional feature spaces, and (3) exposing the models to a Gaussian noise stress-test to evaluate their resilience to the visual artifacts frequently encountered in real-world deployment.

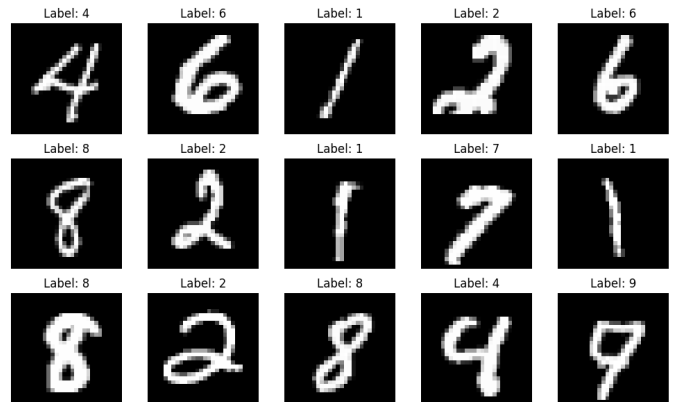


Fig. 1: Representative MNIST Handwritten Digit Samples: A grid illustrating the significant intra-class variability in stroke width, slant, and curvature that classification algorithms must mathematically overcome.

II. RELATED WORK

The classification of handwritten digits has served as a historical testing ground for both classical machine learning and modern deep neural networks.

Initial breakthroughs in high-accuracy classification were driven by Support Vector Machines. Cortes and Vapnik [2] introduced the modern SVM architecture, demonstrating that non-linear decision boundaries could be computed efficiently using the kernel trick. To manage the computational scaling of SVMs on high-dimensional image data, Principal Component Analysis (PCA) [3] became the standard preprocessing step. Furthermore, visualization techniques such as t-SNE, developed by van der Maaten and Hinton [4], allowed researchers to empirically verify the separability of compressed feature spaces before classification.

Despite the mathematical robustness of SVMs, the field experienced a paradigm shift toward deep learning with the introduction of Convolutional Neural Networks (CNNs). LeCun et al. [1] pioneered this transition by applying gradient-based learning to document recognition, proving that convolutional layers could automatically extract spatial hierarchies directly from raw 2D pixels, eliminating the need for manual feature engineering.

Subsequent research focused on stabilizing and optimizing these deep architectures for larger datasets. Srivastava et al. [5] introduced Dropout regularization to prevent the co-adaptation of hidden units, effectively mitigating the rampant overfitting observed in early, unregularized neural networks. Additionally, the development of the Adam optimizer by Kingma and Ba [6] provided computationally efficient, adaptive learning rates, quickly becoming the standard for training deep models.

Building upon these foundational studies, this paper conducts an end-to-end comparative analysis. Unlike prior studies that focus solely on classification accuracy, this work additionally evaluates robustness under Gaussian noise and compares computational scalability between classical and deep learning approaches.

III. THEORETICAL BACKGROUND

To achieve state-of-the-art accuracy, this project moves beyond standard library implementations by mathematically optimizing the core algorithms.

A. The Necessity of the Kernel Trick in SVMs

Support Vector Machines aim to find an optimal hyperplane that separates data classes with the maximum margin. In a linearly separable scenario, this boundary is a flat line or plane in the original spatial dimensions.

However, in raw pixel space, handwritten digits are notoriously non-linearly separable. For instance, an ambiguous "3" and an "8" share an overwhelming majority of active overlapping pixels. A standard linear boundary cannot effectively separate the complex, overlapping geometric manifolds of these digits.

To resolve this, we employ the **Kernel Trick**. Instead of calculating complex boundaries in the original space, a kernel

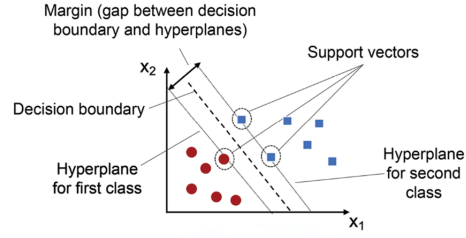


Fig. 2: Visual representation of a baseline Linear SVM maximizing the margin between distinctly separable classes in 2D space.

function implicitly maps the 784-dimensional input vectors into a vastly higher-dimensional feature space where a flat, linear hyperplane can separate the overlapping classes.

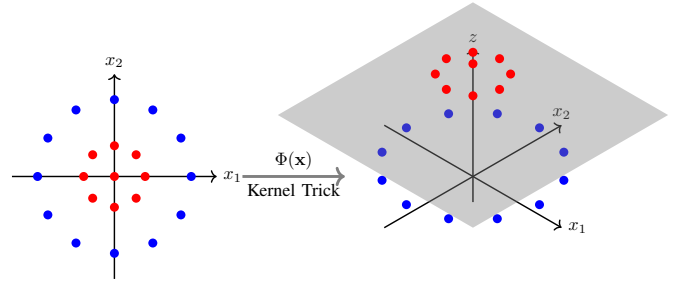


Fig. 3: Visual representation of the Kernel Trick. Non-linearly separable overlapping data (left) is mathematically projected into a higher-dimensional feature space (right) where a linear hyperplane can separate the classes.

Different kernels serve different mathematical purposes:

- **Polynomial Kernels** map the global, broad-stroke geometry of the input vectors. They are excellent for understanding the macro-shape of a digit.
- **RBF (Gaussian) Kernels** measure the localized Euclidean distance between specific pixels. They excel at mapping micro-variations and fine edges.

Mathematical Validity of the Hybrid Kernel:

Our project engineers a **Custom Hybrid Kernel** that blends both equations:

$$K_{hyb}(X, Y) = \alpha \cdot K_{poly}(X, Y) + \beta \cdot K_{rbf}(X, Y) \quad (1)$$

The mathematical validity of this approach is strictly governed by **Mercer's Theorem** and the closure properties of kernel functions. According to kernel theory, if K_1 and K_2 are valid positive semi-definite kernels, then any linear combination $\alpha K_1 + \beta K_2$ (where $\alpha, \beta \geq 0$) is mathematically guaranteed to also be a valid positive semi-definite kernel.

Because we use positive scalars (0.7 and 0.3), our hybrid equation is proven to implicitly map to a true high-dimensional Reproducing Kernel Hilbert Space (RKHS). This ensures that the SVM's convex optimization problem remains mathematically sound, guaranteeing convergence to a global optimum.

while simultaneously evaluating both the global macro-shape and local pixel density of the digits.

B. Spatial Hierarchies in CNNs

Unlike SVMs, which require images to be flattened into an unstructured 1D array, Convolutional Neural Networks (CNNs) process the 2D pixel matrix directly. Convolutions slide mathematical filters across the image to detect primitive features (like vertical or horizontal edges). Subsequent Pooling layers downsample these features, providing *translation invariance*—meaning the network recognizes a "7" whether it is drawn in the center of the image or shifted to the top right. This inherent spatial awareness makes CNNs theoretically superior for image classification.

IV. PROPOSED METHODOLOGY

The experimental framework is organized as a sequential four-notebook pipeline. A shared Google Drive directory acts as the central data store, passing preprocessed tensors and serialized models between stages.

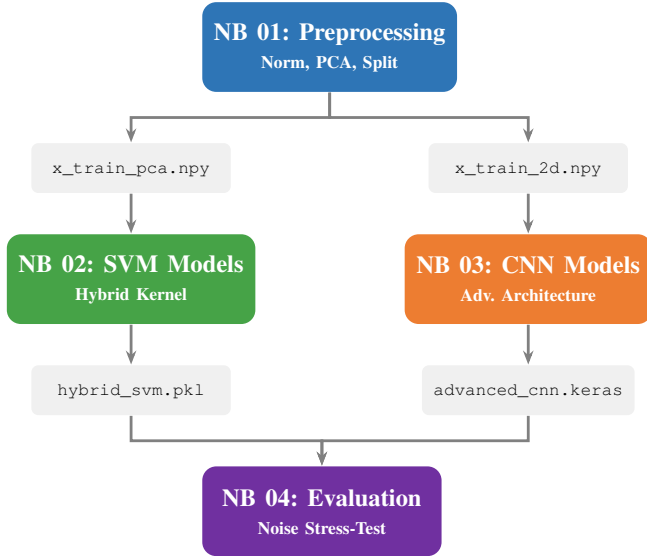


Fig. 4: Complete Hierarchical Data Flow Pipeline. Notebook 01 generates the requisite numpy arrays, splitting them into two distinct mathematical paths. PCA-compressed 1D data flows exclusively to the SVM, while raw 2D spatial tensors flow exclusively to the CNN. Final models are then aggregated in Notebook 04.

A. Dimensionality Reduction (PCA & t-SNE)

Raw image classification with non-linear SVMs is computationally expensive, as the pairwise distance matrix scales quadratically $O(n^2)$ with the number of samples and dimensions. To resolve this, we applied Principal Component Analysis (PCA) parameterized to retain exactly 85% of the cumulative dataset variance. This dynamically compressed the feature space from 784 dimensions down to just 59 components.

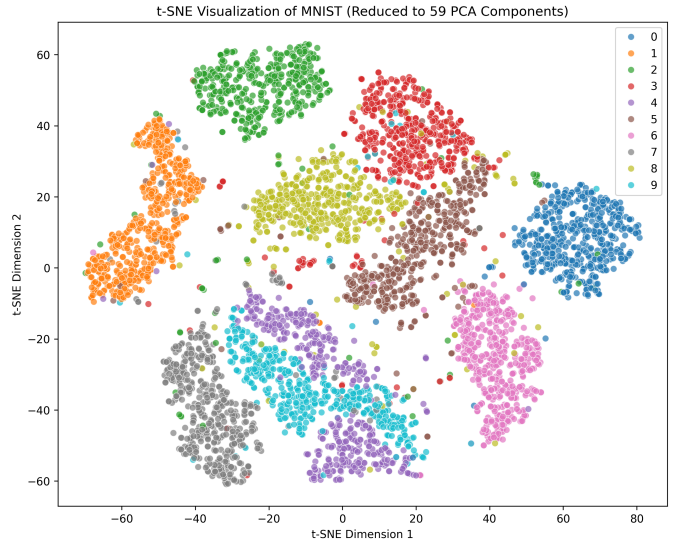


Fig. 5: t-SNE Visualization mapping the 59 PCA components into 2D space. The presence of distinct, well-separated clusters confirms that our aggressive compression mathematically preserved the variance necessary for accurate classification.

B. Detailed SVM Optimization Pipeline

Because SVMs are strictly binary classifiers, natively categorizing 10 separate digits requires a One-vs-One (OvO) ensemble strategy, constructing 45 distinct binary models ($\frac{10 \times 9}{2}$).

During our ablation study on the PCA data, we engineered the custom hybrid kernel defined as:

$$K_{hyb}(X, Y) = 0.7 \cdot K_{poly}(X, Y) + 0.3 \cdot K_{rbf}(X, Y) \quad (2)$$

We deliberately weighted the polynomial component heavier (70%) because the overarching structural integrity of a digit is generally more discriminative than localized pixel noise. The RBF component (30%) acts as a fine-tuning mechanism for complex edge cases.

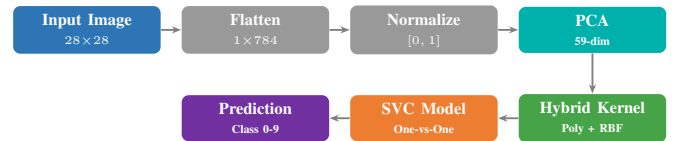


Fig. 6: Detailed SVM Processing Pipeline. Raw images are flattened, normalized, and compressed via PCA before passing through the mathematically blended Hybrid Kernel and the OvO SVM ensemble.

C. Evolution of the CNN Architecture

Deep learning models require careful architectural tuning to prevent memorization of the training data. We developed two distinct CNN pipelines to demonstrate this progression.

1) *The Basic CNN Pipeline:* The Basic CNN serves as our deep learning baseline. It consists of a single Convolutional layer followed immediately by flattening and dense classification.



Fig. 7: Basic CNN Architecture. Without spatial downsampling (Pooling) or regularization (Dropout), this shallow network is highly susceptible to overfitting the training data.

While the Basic CNN trains rapidly, the lack of MaxPooling means the network fails to learn translation invariance, and the lack of Dropout means it relies too heavily on specific pixel activations, leading to divergence between training and validation accuracy.

2) *The Advanced CNN Pipeline:* To resolve the limitations of the basic model, the Advanced CNN implements a dual-block feature extraction architecture before flattening into the dense classification layer.

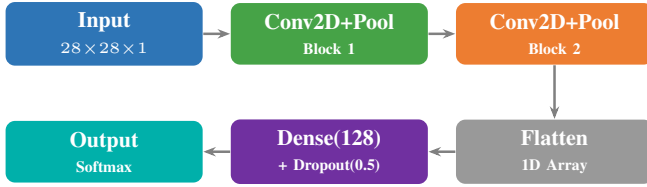


Fig. 8: Detailed Advanced CNN Architecture. The addition of dual MaxPooling layers forces spatial hierarchy learning, while explicitly Flattening into a 50% Dropout layer helps reduce overfitting prior to final Softmax evaluation.

V. IMPLEMENTATION DETAILS

The implementation utilizes Google Colab’s cloud computing environment with GPU acceleration. The core algorithmic stack includes **TensorFlow/Keras** for deep learning model construction, **scikit-learn** for PCA dimensionality reduction and SVC execution, and **Matplotlib/Seaborn** for evaluation visuals.

VI. EXPERIMENTAL RESULTS

A. Baseline Algorithmic Scaling & Justification

Before applying advanced optimizations, we evaluated the baseline performance of both architectures to empirically justify our required engineering decisions.

First, we tested a standard linear SVM on raw 784-dimensional pixels.

TABLE I: SVM Baseline Scaling (Linear Kernel, Raw Pixels)

Dataset Size	Training Time (s)	Test Acc.
5,000	5.30	91.01%
15,000	13.55	92.19%

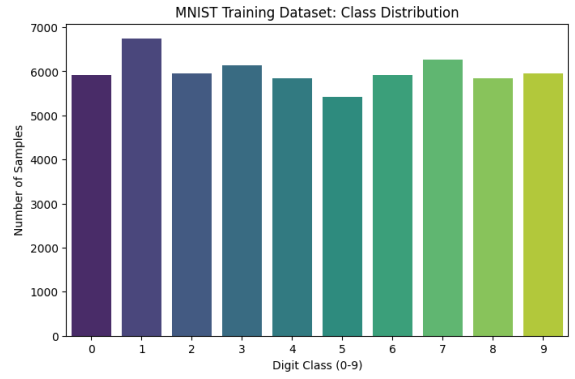


Fig. 9: MNIST Training Set Class Distribution. The dataset is well-balanced, preventing the models from developing a baseline frequency bias toward any specific digit.

As shown in Table I, tripling the dataset size resulted in a $\sim 2.5\times$ increase in computational training time but yielded only a marginal 1.18% improvement in test accuracy. This rapid accuracy plateau proved that the digits were not linearly separable, forcing the integration of PCA compression and Non-Linear mapping.

Next, we evaluated the basic, unregularized CNN architecture.

TABLE II: CNN Baseline Scaling (No Regularization)

Dataset Size	Train Time	Train Acc.	Test Acc.
5,000	12.47s	98.30%	94.79%
15,000	14.23s	99.33%	97.41%

While the CNN parallel-processed the data efficiently without scaling bottlenecks, Table II reveals a critical structural flaw. On the 15,000-sample split, the model exhibited severe overfitting, with the training accuracy diverging from the test accuracy by nearly 2%. This empirically mandated the integration of Dropout and MaxPooling for the advanced model.

B. SVM Kernel Ablation Study

Due to the extreme memory allocation required to compute $O(n^2)$ pairwise distance matrices for the SVM, the kernel ablation study was executed on a randomized 15,000-sample PCA subset to prevent hardware memory overflow.

TABLE III: SVM Ablation Study (15k PCA Subset)

Kernel Configuration	Feature Space	Test Acc.
Linear	PCA (59 dims)	93.11%
Polynomial (degree=3)	PCA (59 dims)	97.23%
Hybrid (0.7 Poly + 0.3 RBF)	PCA (59 dims)	97.68%

The custom Hybrid Kernel definitively outperformed both the Linear and standard Polynomial configurations, demonstrating that blending localized RBF mapping with global

polynomial shapes mathematically optimizes classical model performance.

C. Advanced CNN Performance

The advanced CNN completely bypassed the $O(n^2)$ memory bottlenecks by utilizing isolated mini-batches (batch size = 64) during gradient descent. This allowed us to train the model on the entire 60,000-image dataset over 10 epochs.

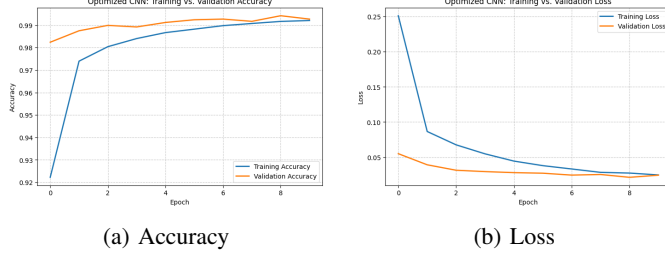


Fig. 10: Advanced CNN Training Curves. The validation metrics closely track the training metrics, proving that our Max-Pooling and Dropout regularization successfully prevented overfitting.

The regularization upgrades were entirely successful. The Advanced CNN achieved a final test accuracy of **99.29%**.

D. Error Analysis via Confusion Matrices

To understand the logical blind spots of our models, we generated confusion matrices based on their clean data predictions.

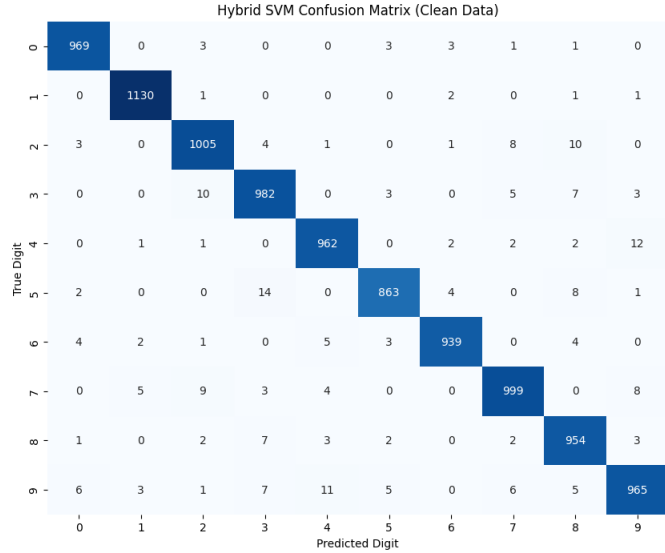


Fig. 11: Hybrid SVM Confusion Matrix. The SVM exhibits clusters of structural false positives, frequently confusing structurally similar digits like 4s and 9s, or 3s and 8s.

E. Computational Scaling Comparison

A key differentiator between classical and deep learning architectures is computational scalability.

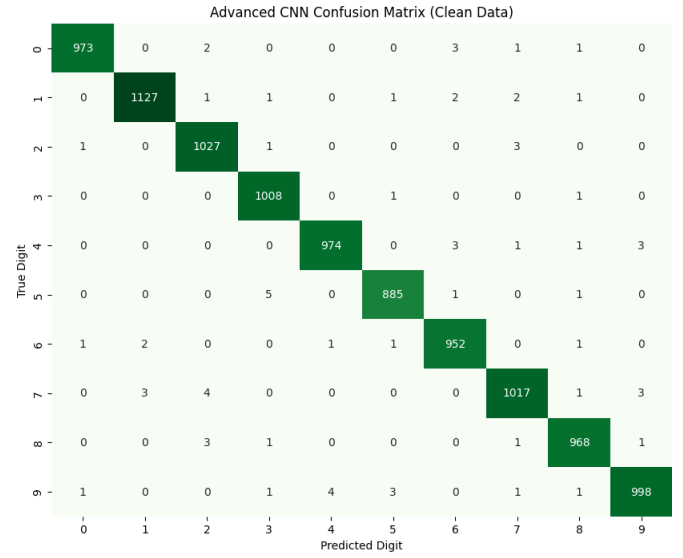


Fig. 12: Advanced CNN Confusion Matrix. The CNN exhibits near-perfect diagonal dominance. By learning specific spatial hierarchies (e.g., detecting horizontal crossbars intersecting vertical strokes), it easily distinguishes subtle handwriting variations that confused the SVM.

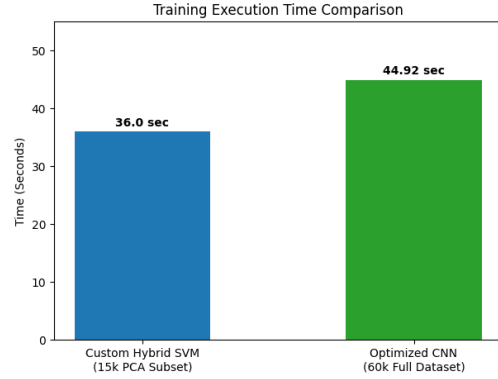


Fig. 13: Training Execution Time Comparison. The CNN is significantly more scalable due to tensor-based parallel processing.

The CNN processes the entire 60,000-image dataset in roughly the same time (44.92s) it takes the SVM to process a heavily compressed 15,000-image subset (36.0s). It is important to note that both models were trained utilizing GPU acceleration. This perfectly illustrates a fundamental algorithmic constraint: while the CNN scales highly efficiently across massive datasets using mini-batch tensor operations, the SVM remains mathematically bound by the $O(n^2)$ complexity of calculating the pairwise distance matrix for its non-linear kernel. Consequently, even with equivalent high-performance hardware, the SVM experiences severe scaling bottlenecks, highlighting the structural scalability advantages of modern deep learning frameworks.

F. Robustness Evaluation Under Gaussian Noise

To determine which architecture is truly superior for real-world deployment, both final models were evaluated under a custom Gaussian noise stress-test ($\sigma = 0.3$), simulating scanner artifacts or poor camera quality.

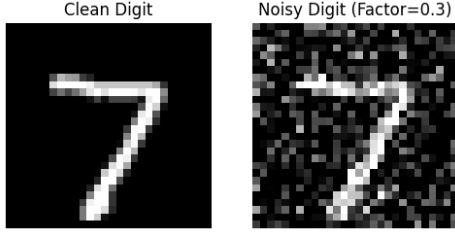


Fig. 14: Gaussian Noise Corruption Example. Clean digit (left) vs. Corrupted digit (right) simulating real-world sensor artifacts.

TABLE IV: Clean vs. Noisy Performance Degradation

Model	Clean Acc.	Noisy Acc.	Drop (Δ)
Hybrid SVM (15k)	97.68%	81.00%	-16.68%
Advanced CNN (60k)	99.29%	95.59%	-3.70%

The stress test demonstrated the architectural superiority of Deep Learning for image processing. The SVM suffered a significant 16.68% drop in accuracy. This occurs because the SVM relies on PCA mapping, and introducing Gaussian noise significantly alters the pixel distribution, pushing data points across the strict mathematical hyperplanes. Conversely, the CNN demonstrated remarkable resilience, suffering only a negligible 3.70% degradation. The CNN’s pooling layers help suppress localized noise while preserving dominant structural features, effectively focusing solely on the core shape of the digit.

VII. CONCLUSION

This paper presented a comprehensive, pipeline-driven comparative study of handwritten digit recognition. By mathematically engineering a custom Hybrid Kernel and utilizing PCA dimensionality reduction, we pushed traditional Support Vector Machines to a highly respectable 97.68% accuracy limit. However, the classical algorithm fundamentally lacked the flexibility required to handle real-world visual noise.

The regularized Convolutional Neural Network outperformed the SVM across most evaluation metrics. It scaled exponentially better to massive datasets (processing 60,000 images in under 45 seconds) and achieved a near-perfect 99.29% test accuracy. Furthermore, its hierarchical feature extraction and spatial pooling proved significantly better at filtering out visual artifacts during our Gaussian noise stress-test, establishing the CNN as a more suitable architecture for practical image recognition systems.

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